

SMART INDIA HACKATHON 2026



- **Problem Statement ID:** SIH26236
- **Problem Statement Title:** AI-Based Intelligent Food Packaging Material Recommendation System for Food Commodities
- **Theme:** Agriculture, FoodTech & Rural Development
- **PS Category:** Software
- **Team ID:** 137649
- **Team Name:** Kalpataru



PROPOSED SOLUTION:

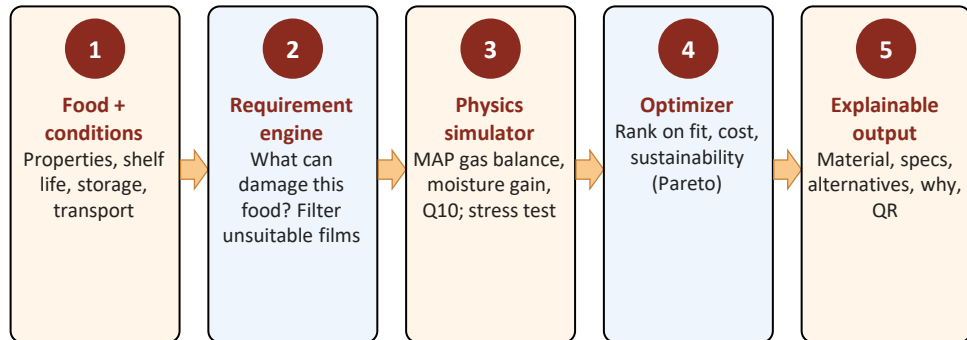
PackPulse does not look up a film. It simulates what happens inside the pack, then picks the cheapest, greenest film that keeps the food safe.

- Inputs: commodity, moisture, fat, pH, respiration rate, shelf life, temperature, RH, transport, storage type
- Computes required OTR, WVTR, thickness, sealability, strength, MAP suitability and gas mix
- Ranks LDPE, HDPE, PET, metalized, foil laminate, biodegradable and breathable films on fit, cost and sustainability
- Explains why, with source and confidence for every number

HOW IT ADDRESSES THE PROBLEM :

- ▶ **Expertise gap:** farmers, MSMEs and startups get expert-level packaging advice instantly, in regional languages, without consultants or lab trials.
- ▶ **Food loss:** requirements come from moisture-gain and respiration physics, preventing moisture pickup, rancidity, mould and “suffocated” produce.
- ▶ **Over-packaging:** finds the minimum sufficient barrier, cutting packaging cost and plastic waste.

WORKFLOW DIAGRAM:



Example: potato chips, ambient, 6 months → high O₂, moisture and light barrier → computed OTR/WVTR limits → metalized PET/PE laminate, N₂-flush MAP, plus cheaper and greener alternatives.

INNOVATION AND UNIQUENESS OF THE SOLUTION :

- ▶ **Physics-informed:** computes the required OTR/WVTR instead of guessing (MAP equilibrium, Labuza moisture gain, Q10).
- ▶ **Pack digital twin:** Monte Carlo supply-chain stress test gives the probability the food survives its shelf life.
- ▶ **Right-sizing + traceable data:** Pareto options (best / cheapest / greenest); every number carries source, confidence or a “data gap” flag.

TECH STACK:

.js 16 (React) with
Glassmorphic UI; Vernacular search
(Hindi / regional aliases); Mobile-first
Backend / API: FastAPI (Python 3.11+),
Pydantic v2 validation, Uvicorn high-
throughput ASGI server

Physics engine: SciPy (ODE) for MAP gas balance; Labuza moisture-gain model;
Q10 / Arrhenius temperature correction

ML ranker: scikit-learn / XGBoost and similarity search on the normalized
material dataset

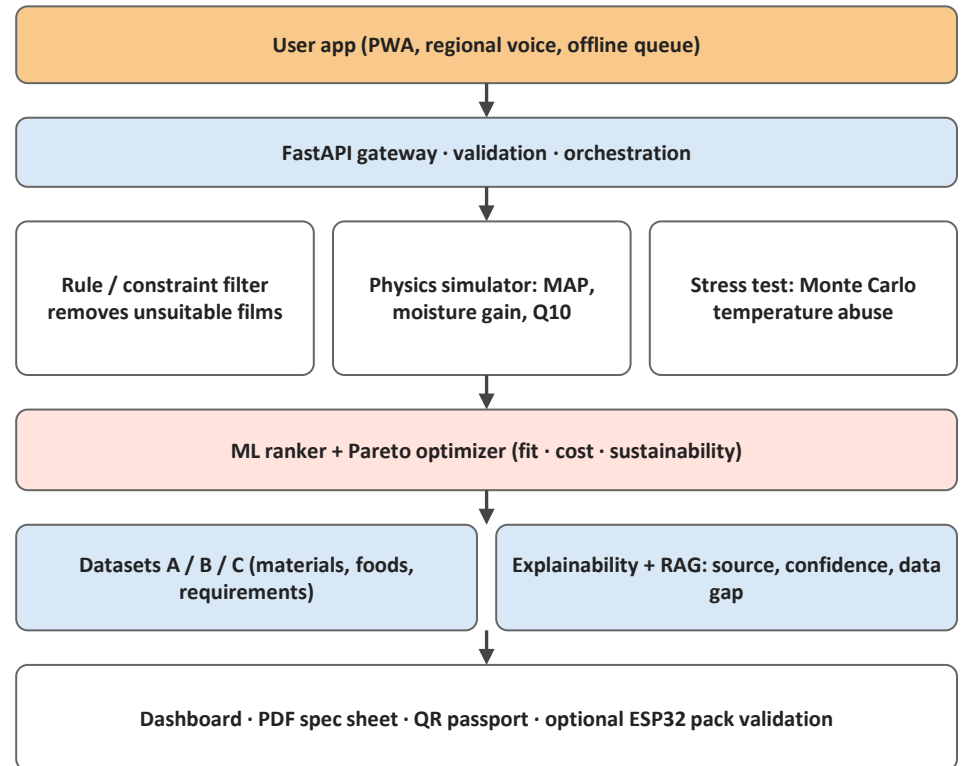
Optimizer: NSGA-II or weighted Pareto over technical fit, cost, CO₂ footprint and
recyclability

Data: PostgreSQL / SQLite: materials, foods, requirement mapping, each row
with source, DOI and confidence

Output / modules: Dynamic Pouch

Costing & MOQ Simulator, Digital
Packaging Passport (DPP) with QR
generation, Europe PMC Crawler

ARCHITECTURE DIAGRAM :



TECHNICAL FEASIBILITY

Uses established food-packaging science (MAP modelling, Labuza shelf life) with open-source Python tools; ML only ranks options.

ECONOMIC FEASIBILITY

Open datasets and open-source stack keep build and hosting cost low; no special hardware needed for users.

USER VIABILITY

Simple form or voice in Hindi/Marathi; plain-language “why” lets non-experts trust and act on the result.

OPERATIONAL FEASIBILITY

Starts with 20–30 commodities; experts from NIFTEM/CFTRI-type institutes can validate the requirement tables.

SCALABILITY

Add foods, films and languages as data rows; modular services and an API serve FPOs and cold-chain firms.

FINANCIAL VIABILITY

Free tier for farmers; subscriptions and API access for MSMEs and industry (see Impact slide).

Challenges	Strategy
Credible packaging dataset	Tiered sources (Food-Pack-Mapper, CIRAD, NIFTEM-T, CFTRI, FAO); confidence tag and DOI on every row.
No ground truth for ML	Requirements come from physics; ML is used only for ranking and similarity.
Mixed units and test conditions	Normalize OTR/WVTR units, temperature and RH before use; never mix raw values.
Missing data for new foods	Show a “data gap” flag instead of guessing; route to expert review.

Model	Purpose	Key inputs
MAP gas balance	Required OTR, micro-perforation, gas mix	Respiration rate, mass, area, target O_2/CO_2
Labuza moisture gain	Maximum allowed WVTR	Initial, critical, equilibrium moisture
Q10 / Arrhenius	Temperature correction	Storage and transport profile
Monte Carlo stress test	Survival probability	Temperature-abuse scenarios
Pareto optimizer	Best / cheapest / greenest	Fit, cost, CO_2 , recyclability

POTENTIAL IMPACT ON THE TARGET AUDIENCE

Farmers

Right post-harvest packaging means less spoilage and better prices.

MSMEs & startups

No expert fees or trial-and-error cost for new products.

Food industry

Right-sized packaging cuts cost, plastic and rejected batches.

Environment

Recyclable options and less over-packaging lower plastic and CO₂.

SELF-SUSTAINING BUSINESS MODEL

- **Free tier:** farmers, FPOs and students get core recommendations at no cost.
- **SaaS plans:** MSMEs and manufacturers pay for saved projects, stress tests and PDF spec sheets.
- **API access:** cold-chain firms, FPOs and government portals integrate PackPulse.
- **Supplier leads:** verified film suppliers pay for qualified enquiries, kept separate from ranking.
- **Partnerships:** MoFPI, NIFTEM and CSR grants for rural onboarding.

Benefits	Value delivered
Expert advice for everyone	Food properties in, packaging decision out, no consultant needed.
Complete specifications	OTR, WVTR, thickness, sealability, strength, MAP suitability.
Fresh-produce intelligence	Respiration-aware breathable / micro-perforated film and gas mix.
Cost optimization	Best film within the user's per-pack budget.
Sustainability analysis	Recyclable and eco-friendly alternatives with footprint score.
Shelf-life and risk prediction	Survival probability across the real supply chain.
QR traceability	Pack spec, storage advice and recycling instructions in one QR.

FUTURE SCOPE

- Smart Pack Validator: low-cost ESP32 O₂/CO₂/RH sensor to compare real and simulated gas curves and recalibrate
- Regional-language voice / WhatsApp copilot for farmers
- Supplier marketplace and API for FPOs, cold-chain firms and government portals
- Expansion beyond 30 commodities as verified data grows

RESEARCH VALIDATION

- **Food-Pack-Mapper (npj Science of Food, 2026):** 295 OTR/WVTR data points from 49 studies, standardized units; open dataset and code.
- **CIRAD Dataverse:** food-packaging composition and gas-permeability data from 50 articles, annotated by three experts.
- **Indian sources:** NIFTEM-T packaging-resin handbook and CFTRI packaging studies add WVTR, strength and sealing data.
- **FAO packaging properties table:** thickness, transmission, strength and sealing temperature for common films.
- **Models:** Labuza moisture-gain shelf-life model and MAP respiration modelling. Add DOIs for every source used.

Physics for requirements, literature data for materials, a DOI on every number.

COMPETITIVE FEATURE GAP ANALYSIS

What PS 26236 demands	PackPulse	Consultants / lab trials	Rule-based apps	Black-box ML
Food-property based input	✓	✓	△	✓
Computed OTR / WVTR / thickness specs	✓	✓	✗	△
Respiration-aware MAP and micro-perforation	✓	✓	✗	△
Cost + sustainability optimization	✓	△	✗	△
Explainable, source-grounded output	✓	✓	△	✗
Regional-language, farmer-friendly access	✓	✗	✗	✗
QR traceability	✓	✗	✗	✗

✓ Available | △ Partial / limited | ✗ Not a core offering (team's indicative assessment)